

server machine and boot it up. If your machine is not set to boot from a CD/USB, you may need to press F12 to enter a one-time boot menu and select CD/USB, or you could change the boot device priority in the BIOS settings (but make sure you change it back if you are going to use external disks for storage—which is the likely scenario if you are using an old laptop for your server, like I am). Your system's start-up screen will tell you how to enter the BIOS set-up screen.

After booting into Ubuntu, select the Install Ubuntu option and follow the onscreen instructions to set up a dual-boot or single-boot system, based on your preferences. Also, make sure you select log in automatically when you are creating a user account, so that you have a single-button power-up sequence, which doesn't need any action from you during start-up.

The server set-up

At this point, I assume you have Ubuntu set up successfully and are booted into it. Now for the server set-up, plug in your external drives if you haven't already, and click Ubuntu Dash (the button with Ubuntu logo) and search for Terminal. Ubuntu has built-in support for NTFS, so your drives should ideally work out-of-the-box. If you are configuring new drives, you can format them as ext4 for better Linux compatibility. Ubuntu supports HFS-formatted disks (the native format on Macintosh computers) as well, as long as journaling is disabled on them; but for the purposes of this article, let us assume your disks are formatted as NTFS or ext4.

Step 1. Mount-points: In the terminal, run the following commands as explained below:

```
sudo mkdir /path/to/mountpoint
sudo chown sumit /path/to/mountpoint
```

Replace /path/to/mountpoint with a folder in which you would like to mount your hard disk. For example:

```
sudo mkdir /media/BackupDisk
sudo chown sumit /media/BackupDisk
```

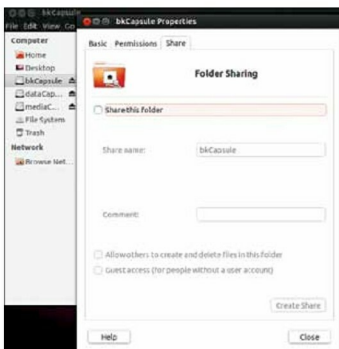


Fig. 1. Sharing options

```
sudo mkdir /media/MovieDisk
sudo chown sumit /media/MovieDisk
```

Enter your password when you are prompted for it. Repeat the commands for each hard disk you have connected to the system.

Step 2. Auto-mounting: Next, run the following command in the terminal:

```
sudo blkid
```

This command should result in an output that looks something like what follows:

```
/dev/sda1: TYPE="ntfs"
        UUID="A0F0582EF0580CC2"
/dev/sda2: UUID="8c2da865-13f4-47a2-9c92-2f31738469e8"
        SEC_TYPE="ext2" TYPE="ext3"
/dev/sda3: TYPE="swap"
        UUID="5641913f-9bcc-4d8a-8bcb-ddfc3159e70f"
/dev/sda5: UUID="FAB008D6B0089AF1"
        TYPE="ntfs"
```

Basically, what we are trying to do here is to make sure your disks mount automatically every time you boot up your server. The fstab or file systems table is a configuration file in Linux systems that lists connected disks and partitions and their initialisation/mounting information. This information is used by the mount tool to mount your disks/partitions. So we need to start adding entries for each hard disk into the /

etc/fstab file. We could use GUI tools, but those would add the disks using their device file names (/dev/sdX). This leads to problems in some cases, when the system renames these device files, causing errors while mounting disks. That is where Universal Unique Identifier (UUID) comes to the rescue. UUID enables you to uniquely identify your disk; and it does not change with every reboot, like the device files do. UUID is conceptually similar to the MAC address for a

network card.

In the terminal, run `sudo gedit /etc/fstab` to open a text editor window (like Notepad on Windows). Add an entry to the end of the file in the following format, for each of your connected disks' partitions:

```
UUID=Your-UUID /path/to/mountpoint
fs-type defaults 0 0
```

You will need to copy the UUID from the list generated by the `blkid` command. Here /path/to/mountpoint is the address to the directories you created in Step 1. For example:

```
UUID=A0F0582EF0580CC2 /media/BackupDisk ntfs defaults 0 0
UUID=FAB008D6B0089AF1 /media/MovieDisk ntfs defaults 0 0
```

Now save and close this file. You should be able to see all your disks in the left-hand sidebar of your file explorer (Nautilus). However, these disks are not mounted yet. You can click them individually to mount them in the folder you have entered in your fstab file (/media/ in our case). The next time you boot up your machine, this should happen automatically.

Sharing your drives on the network

Now that the basic set-up is out of the way, and your disks are usable by the server, you need to get down to some